

SUMMARY AND RECOMMENDATIONS

Telecom customer churn: 1,869 of 7,043 customers churned (26.5% overall)

◆ Key Insights by Payment Method

1. Electronic Check

- Churned: **1071 (45.3%)**
- Retained: **1294 (54.7%)**
- 📌 *Highest raw churn rate of all methods. Customers using electronic checks are almost equally likely to churn or stay, showing a high-risk group. Part of this comes from contract type: **78%** of e-check users are on month-to-month contracts. Among month-to-month customers, e-check churns at **53.7%** vs ~**33%** for auto-pay.*

2. Mailed Check

- Churned: **308 (19.1%)**
- Retained: **1304 (80.9%)**
- 📌 *Much lower raw churn than electronic checks, but this partly reflects contract mix (55% month-to-month). Within month-to-month contracts, mailed-check churn (**31.6%**) is similar to auto-pay, so this is not a high-risk group.*

3. Bank Transfer (Automatic)

- Churned: **258 (16.7%)**
- Retained: **1286 (83.3%)**
- 📌 *Low raw churn rate. Part of this reflects contract mix (38% month-to-month); within month-to-month contracts churn is **34.1%**.*

4. Credit Card (Automatic)

- Churned: **232 (15.2%)**
- Retained: **1290 (84.8%)**
- 📌 *Lowest raw churn rate of all methods, but this also partly reflects contract mix (36% month-to-month). Within month-to-month contracts churn is **32.8%**, similar to bank transfer and mailed check.*

◆ Is the Electronic Check Effect Real? (Controlling for Contract)

Payment method and contract type overlap, so raw churn rates by payment method overstate the payment effect. Churn rate by payment method **within** each contract type:

Contract	Bank transfer (auto)	Credit card (auto)	Electronic check	Mailed check
Month-to-month	34.1%	32.8%	53.7%	31.6%
One year	9.7%	10.3%	18.4%	6.8%
Two year	3.4%	2.2%	7.7%	0.8%

Two-year electronic check group is small (168 customers); read that cell with caution.

- **Contract type is the strongest driver:** churn is **42.7%** on month-to-month, **11.3%** on one-year and **2.8%** on two-year contracts. **88.6%** of all churners are month-to-month.
- Electronic check has the highest churn in **every** contract type, so payment method adds real risk on top of contract.
- The gap is smaller than the raw numbers suggest: about **20 points** on month-to-month (53.7% vs ~33.5% for auto-pay), compared with ~29 points in the raw figures (45.3% vs ~16%).

◆ Other Churn Drivers

Churn rate within each group (overall churn is 26.5%). Internet-related rows compare customers with internet service only, where noted.

Driver	Higher-churn group	Churn	Comparison group	Churn
Tenure	0–6 months	52.9%	49–72 months	9.5%
Contract	Month-to-month	42.7%	Two year	2.8%
Internet service	Fiber optic	41.9%	DSL	19.0%
Security + Tech Support*	Neither	49.0%	Both	9.0%
Senior citizen	Yes	41.7%	No	23.6%
Paperless billing	Yes	33.6%	No	16.3%
Partner	No	33.0%	Yes	19.7%
Dependents	No	31.3%	Yes	15.5%
Gender	Female	26.9%	Male	26.2%

**Internet customers only: 2,553 have neither Online Security nor Tech Support, 1,099 have both.*

- **55%** of churners leave within their first 12 months, and churn falls steadily with tenure.
- Customers without Online Security and Tech Support churn at **49.0%** vs **9.0%** for those with both. Streaming TV/Movies and phone service show little difference.
- Churned customers pay more per month (**74.4** vs **61.3** on average) and account for **30.5%** of total monthly charges.
- Gender has no meaningful effect (26.9% vs 26.2%).
- **Caution:** these are associations, not causes. Tenure, contract, internet service and payment method overlap heavily. For example, the Fiber optic gap may reflect price or contract mix rather than the technology.

◆ Recommendations

1. Move Month-to-Month Customers to Longer Contracts

- Month-to-month churn is **42.7%** vs **11.3%** (one year) and **2.8%** (two year), and **89%** of all churners are month-to-month.
- Offer discounts or perks for **1–2 year commitments**. Contract type is a bigger churn driver than payment method.

2. Focus Retention on the First 6–12 Months

- Churn is **52.9%** in the first 6 months vs **9.5%** after 4 years, and 55% of churners leave within a year.
- Invest in **onboarding, early check-ins and first-year offers** for new customers.

3. Target Electronic Check Customers on Month-to-Month Contracts

- Focus on the **1,850** e-check customers on month-to-month contracts (**53.7% churn**), the highest-risk segment.
- Promote **Credit Card or Bank Transfer Auto-pay** with incentives such as discounts, reward points or a free month, and explain the convenience and security of automatic payments by email or in-app messages.

4. Promote Online Security and Tech Support

- Internet customers with neither service churn at **49.0%** vs **9.0%** for those with both.
- Bundle or offer these services to internet customers, especially **Fiber optic** users (41.9% churn).

5. Test Before Scaling

- Run an **A/B test of an auto-pay incentive**, since this analysis cannot show that payment method itself causes churn.

6. Monitor Churn Regularly

- Set up **dashboards/KPIs** to monitor churn by contract, tenure and payment method.
- This helps track whether retention actions are working.

Main takeaway:

Churn is concentrated among **new customers on month-to-month contracts**: contract type and tenure are the strongest drivers, and lack of security/tech-support services adds further risk. Electronic check adds extra risk on top, but about 20 points on month-to-month rather than the raw 30. Bringing month-to-month e-check customers down to the auto-pay churn rate (~33.5%) could retain roughly **375 customers, about 20% of total churn**. This assumes the payment-method effect is causal, so an **A/B test of an auto-pay incentive** is the right next step.